

with (LinearAlgebra);

[&x, Add, Adjoint, BackwardSubstitute, BandMatrix, Basis, BezoutMatrix, BidiagonalForm, BilinearForm, CARE, CharacteristicMatrix, CharacteristicPolynomial, Column, ColumnDimension, ColumnOperation, ColumnSpace, CompanionMatrix, CompressedSparseForm, ConditionNumber, ConstantMatrix, ConstantVector, Copy, CreatePermutation, CrossProduct, DARE, DeleteColumn, DeleteRow, Determinant, Diagonal, DiagonalMatrix, Dimension, Dimensions, DotProduct, EigenConditionNumbers, Eigenvalues, Eigenvectors, Equal, ForwardSubstitute, FrobeniusForm, FromCompressedSparseForm, FromSplitForm, GaussianElimination, GenerateEquations, GenerateMatrix, Generic, GetResultDataType, GetResultShape, GivensRotationMatrix, GramSchmidt, HankelMatrix, HermiteForm, HermitianTranspose, HessenbergForm, HilbertMatrix, HouseholderMatrix, IdentityMatrix, IntersectionBasis, IsDefinite, IsOrthogonal, IsSimilar, IsUnitary, JordanBlockMatrix, JordanForm, KroneckerProduct, LA_Main, LUdecomposition, LeastSquares, LinearSolve, LyapunovSolve, Map, Map2, MatrixAdd, MatrixExponential, MatrixFunction, MatrixInverse, MatrixMatrixMultiply, MatrixNorm, MatrixPower, MatrixScalarMultiply, MatrixVectorMultiply, MinimalPolynomial, Minor, Modular, Multiply, NoUserValue, Norm, Normalize, NullSpace, OuterProductMatrix, Permanent, Pivot, PopovForm, ProjectionMatrix, QRdecomposition, RandomMatrix, RandomVector, Rank, RationalCanonicalForm, ReducedRowEchelonForm, Row, RowDimension, RowOperation, RowSpace, ScalarMatrix, ScalarMultiply, ScalarVector, SchurForm, SingularValues, SmithForm, SplitForm, StronglyConnectedBlocks, SubMatrix, SubVector, SumBasis, SylvesterMatrix, SylvesterSolve, ToeplitzMatrix, Trace, Transpose, TridiagonalForm, UnitVector, VandermondeMatrix, VectorAdd, VectorAngle, VectorMatrixMultiply, VectorNorm, VectorScalarMultiply, ZeroMatrix, ZeroVector, Zip]

(1)

with (LinearAlgebra[Generic]);

[BareissAlgorithm, BerkowitzAlgorithm, CharacteristicPolynomial, Determinant, GaussianElimination, GenericCheck, HermiteForm, HessenbergAlgorithm, HessenbergForm, LinearSolve, MatrixAdd, MatrixInverse, MatrixMatrixMultiply, MatrixVectorMultiply, MinorExpansion, NullSpace, RREF, ReducedRowEchelonForm, SmithForm, StronglyConnectedBlocks, VectorAdd]

(2)

$Q[\text{'-'}] := \text{proc}(a, b) \ (a - b) \bmod 101; \text{end proc};$

$Q_{-} := \text{proc}(a, b) \ a - b \bmod 101 \text{ end proc}$ (3)

$Q[\text{'+'}] := \text{proc}(\) \text{ local } x; \text{ add}(x, x = [_passed]) \bmod 101; \text{end proc};$

$Q_{+} := \text{proc}(\) \text{ local } x; \text{ add}(x, x = [args]) \bmod 101 \text{ end proc}$ (4)

$Q[\text{'*'}] := \text{proc}(\) \text{ local } x; \text{ mul}(x, x = [_passed]) \bmod 101; \text{end proc};$

$Q_{*} := \text{proc}(\) \text{ local } x; \text{ mul}(x, x = [args]) \bmod 101 \text{ end proc}$ (5)

$Q[\text{'/'}] := \text{proc}(a, b) \ a / b \bmod 101; \text{end proc};$

$Q_{/} := \text{proc}(a, b) \ a / b \bmod 101 \text{ end proc}$ (6)

$$Q[\text{'='}] := \mathbf{proc}(a, b) \text{ evalb}(a \bmod 101 = b \bmod 101); \mathbf{end proc};$$

$$Q_{\text{'='}} := \mathbf{proc}(a, b) \text{ evalb}(a \bmod 101 = b \bmod 101) \mathbf{end proc} \quad (7)$$

$$Q[\text{'0'}] := 0;$$

$$Q_0 := 0 \quad (8)$$

$$Q[\text{'I'}] := 1;$$

$$Q_I := 1 \quad (9)$$

$$B := \text{Transpose}(\text{ReducedRowEchelonForm}[Q](\text{Transpose}(AA)));$$

$$B := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & 89 & 80 \\ 37 & 97 & 21 \\ 0 & 7 & 69 \\ 62 & 73 & 18 \end{bmatrix} \quad (10)$$

$$\text{ZeroMat} := \text{ZeroMatrix}(1, 3);$$

$$\text{ZeroMat} := \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \quad (11)$$

$$B := \langle \langle B \rangle, \langle \text{ZeroMat} \rangle \rangle;$$

$$B := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & 89 & 80 \\ 37 & 97 & 21 \\ 0 & 7 & 69 \\ 62 & 73 & 18 \\ 0 & 0 & 0 \end{bmatrix} \quad (12)$$

$$b := \text{Matrix}(\begin{bmatrix} [76], [27], [19], [77], [91], [61], [46] \end{bmatrix});$$

$$b := \begin{bmatrix} 76 \\ 27 \\ 19 \\ 77 \\ 91 \\ 61 \\ 46 \end{bmatrix} \quad (13)$$

$$bb := \langle \langle b \rangle, \langle \text{IdentityMatrix}(1) \rangle \rangle;$$

$$bb := \begin{bmatrix} 76 \\ 27 \\ 19 \\ 77 \\ 91 \\ 61 \\ 46 \\ 1 \end{bmatrix} \quad (14)$$

$$B := \langle \langle B | bb \rangle \rangle;$$

$$B := \begin{bmatrix} 1 & 0 & 0 & 76 \\ 0 & 1 & 0 & 27 \\ 0 & 0 & 1 & 19 \\ 3 & 89 & 80 & 77 \\ 37 & 97 & 21 & 91 \\ 0 & 7 & 69 & 61 \\ 62 & 73 & 18 & 46 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (15)$$

$$CC := \langle \langle ZeroMatrix(3, 4) \rangle, \langle 101 * IdentityMatrix(4) \rangle, \langle ZeroMatrix(1, 4) \rangle \rangle;$$

$$CC := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 101 & 0 & 0 & 0 \\ 0 & 101 & 0 & 0 \\ 0 & 0 & 101 & 0 \\ 0 & 0 & 0 & 101 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (16)$$

$$LatticeMatrix := \langle \langle B | CC \rangle \rangle;$$

$$LatticeMatrix := \begin{bmatrix} 1 & 0 & 0 & 76 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 27 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 19 & 0 & 0 & 0 & 0 \\ 3 & 89 & 80 & 77 & 101 & 0 & 0 & 0 \\ 37 & 97 & 21 & 91 & 0 & 101 & 0 & 0 \\ 0 & 7 & 69 & 61 & 0 & 0 & 101 & 0 \\ 62 & 73 & 18 & 46 & 0 & 0 & 0 & 101 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (17)$$

with(IntegerRelations);

$$[LLL, LinearDependency, PSLQ] \quad (18)$$

LLL(Transpose(LatticeMatrix));

$$\begin{bmatrix} 0 & -1 & -1 & -1 & 1 & 0 & 0 & 1 \\ -1 & -3 & 5 & -5 & -3 & -4 & 1 & 1 \\ 4 & 3 & -1 & -1 & 7 & 1 & 4 & -6 \\ -8 & -3 & 7 & -1 & 0 & 8 & -3 & 2 \\ -2 & 9 & 1 & -2 & 1 & 7 & -4 & 5 \\ -8 & 0 & -4 & -5 & -1 & 0 & 7 & -7 \\ -6 & -2 & 2 & 0 & 5 & -4 & -8 & -7 \\ -12 & 1 & 2 & 9 & 5 & -5 & 12 & 6 \end{bmatrix} \quad (19)$$

*AA := Matrix([[64, 6, 77], [57, 89, 82], [24, 3, 91],
[14, 99, 63], [18, 30, 89], [35, 22, 86], [77, 55, 76]]);*

$$AA := \begin{bmatrix} 64 & 6 & 77 \\ 57 & 89 & 82 \\ 24 & 3 & 91 \\ 14 & 99 & 63 \\ 18 & 30 & 89 \\ 35 & 22 & 86 \\ 77 & 55 & 76 \end{bmatrix} \quad (20)$$

ee := Matrix([[0], [-1], [-1], [-1], [1], [0], [0]]);

$$ee := \begin{bmatrix} 0 \\ -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \tag{21}$$

$$As := b - ee;$$

$$As := \begin{bmatrix} 76 \\ 28 \\ 20 \\ 78 \\ 90 \\ 61 \\ 46 \end{bmatrix} \tag{22}$$

$$ss := \text{LinearSolve}[Q](AA, As);$$

$$ss := \begin{bmatrix} 27 \\ 87 \\ 99 \end{bmatrix}, [\] \tag{23}$$

$$e := ee; s := ss;$$

$$e := \begin{bmatrix} 0 \\ -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$s := \begin{bmatrix} 27 \\ 87 \\ 99 \end{bmatrix}, [\] \tag{24}$$